

SkyGuardian

Saving Lives from the Sky

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1 million

UXOs identified in Ukraine so far

75 million

UXOs identified in Laos since the Vietnam War

6,892

UXO casualties in 2019 around the globe



THE NEED



We NEED to create a system to safely identify and physically mark the location of UXOs autonomously in known-mined and hard to reach areas

Gaps in Existing Solutions



Lacking Versatility

- Existing UXO-detection systems only have 1-2 sensors.



Low Accuracy

- False alarm rate of up to 20%



Personnel-Intensive

- Requires one UXO detection engineer
- Requires up to 7 non-UXO personnel in harm's way.

Design Requirements

Identify UXOs

- Use wide array of sensor suite to identify UXOs

Real-Time Data Transmission

- Automatically send UXO coordinates to ground station in real-time.
- Local backup in case connection is lost

Fully-Autonomous

- Drone can operate without any human input.

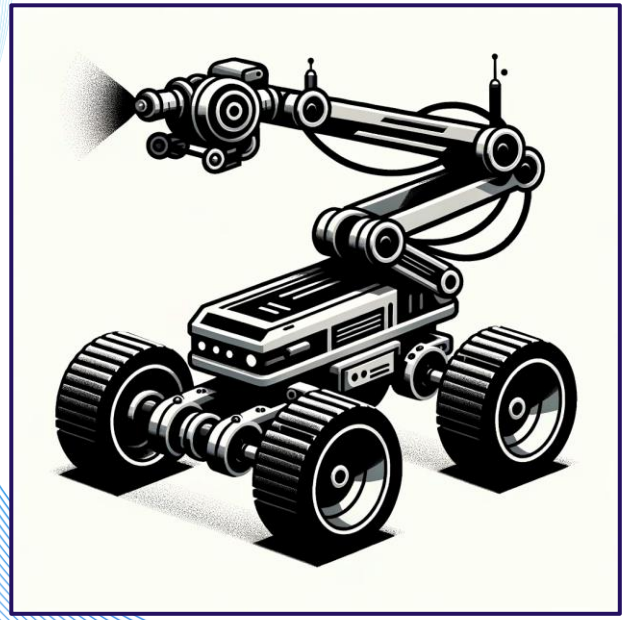
Shielded Electronics

- Protective outer coating

Mark UXOs

- Utilizes dual paint system to physically pinpoint the location of UXO
- Transmits coordinates digitally.

Designs We Considered



Quadcopter Drone

- Less aerodynamic design
- Lower range
- Low payload
- Low stability

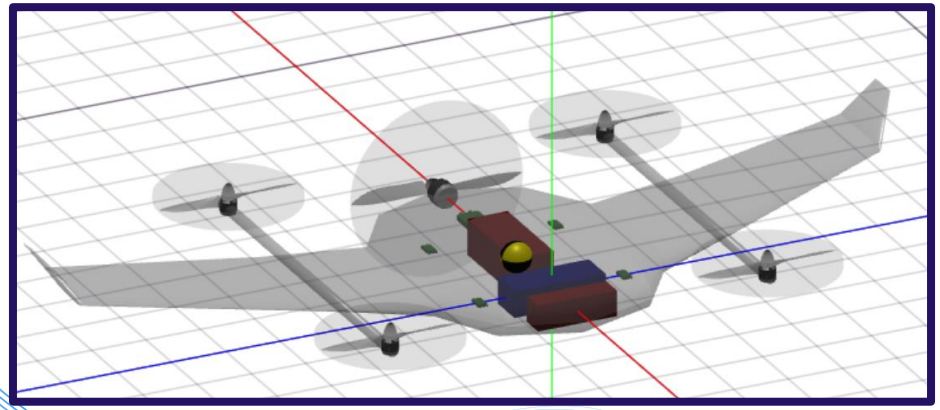


Land-Based Rover

- Weaker signal
- Greater detonation risk

Final Design:
The SkyGuardian

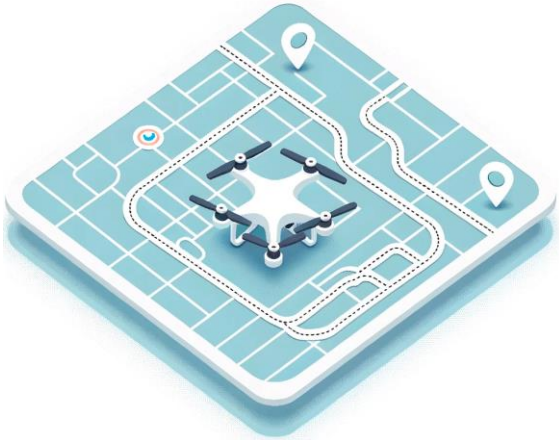
- Added blended wing to increase range
- Deployable payload with UXO detection sensors
- Safer than rover, avoid accidental triggering
- Compact, lightweight design





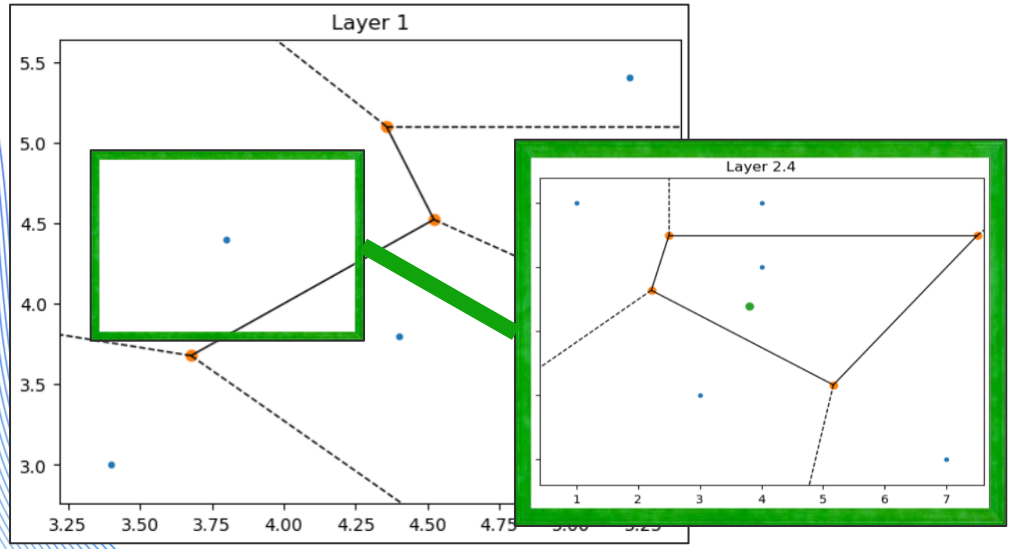
Technical Specifications

- **Range:** 5-10 miles or 2-3 square miles
- **Total scan time:** 30 minutes
- **Speed**
  - Mode 1 : 35-50 mph
  - Mode 2 & 3: 0-5 mph
- **Altitude**
  - Mode 1: 150 feet
  - Mode 2 & 3: 35 feet
  - Drop box module: 2 feet



SkyMap Algorithm

- **Input:** known UXO locations
- **Output:** zones of interest & optimal flight path





Three UXO-Survey Modes



High Altitude Recon

- Sensors: High-res camera and LiDAR
- Cruise Altitude: 150 Feet
- SkyMap algorithm



Low Altitude Scan

- Sensors: Magnetometer
- Altitude: 35 ft
- Hovers over UXOs



Ground Scan

- Sensors: Ground Penetrating Radar, Infrared
- Lowers payload carrying sensors and paint module to the ground

Increasing Precision

Dual-Paint System



Red Paint

Identifies **more urgent**, harmful UXOs:

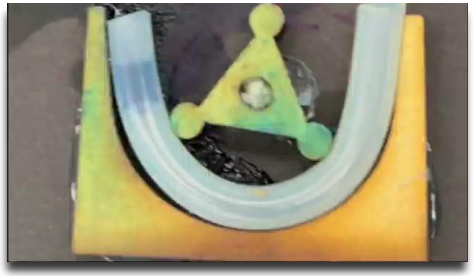
- UXO is in immediate path of citizens/military personnel

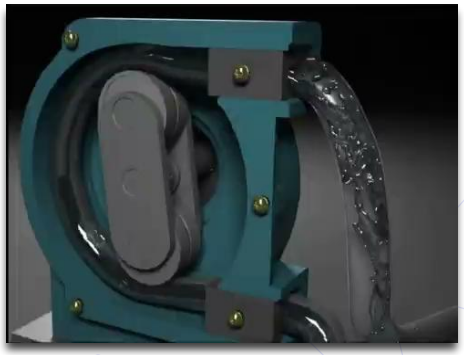
Neon Orange

Identifies **less urgent** UXOs:

- UXO is not in close proximity to personnel

Peristaltic pump





Method of Delivery

- Paint dispenser attached to lowered payload

Paint Characteristics

- Bright Neon
- Non-Sticky
- Long-lasting
- Identifiable

Commercial Viability

DoD UXO Detection Budget

- The US Department of Defense spends about $200 million a year on UXO detection alone.

Only $25,000 per Unit

- Can be produced and deployed at a low cost

Replace up to Seven Personnel

- Perform tasks of up to 7 personnel

In Conclusion

Resources

- https://tamu.libguides.com/c.php?g=1353502&p=9991862
- https://today.umd.edu/unexploded-ordnance-is-scattered-across-ukraines-front-lines-umd-researchers-are-mapping-hot-spots-with-ai
- https://www.youtube.com/watch?app=desktop&v=rtRHmYqR-wQ
- https://www.youtube.com/watch?v=SP99_WMIqns
- https://www.deltaquad.com/vtol-drones/view/
- https://en.wikipedia.org/wiki/Delaunay_triangulation